

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{81} \sum_{\mathbf{p}} g_1^4 L F_{3,0}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{216} \sum_{\mathbf{p}} g_1^4 L F_{4,-1}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{405} \sum_{\mathbf{p}} g_1^4 L F_{5,-2}[m_d^{\mathbf{P}}] \right. \\ & \quad \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} \sum_{\mathbf{p}} g_1^4 L F_{3,0}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{72} \sum_{\mathbf{p}} g_1^4 L F_{4,-1}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{135} \sum_{\mathbf{p}} g_1^4 L F_{5,-2}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{54} \sum_{\mathbf{p}} g_1^4 L F_{3,0}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{144} \sum_{\mathbf{p}} g_1^4 L F_{4,-1}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{135} \sum_{\mathbf{p}} g_1^4 L F_{5,-2}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{162} \sum_{\mathbf{p}} g_1^4 L F_{3,0}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{432} \sum_{\mathbf{p}} g_1^4 L F_{4,-1}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{405} \sum_{\mathbf{p}} g_1^4 L F_{5,-2}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} \sum_{\mathbf{p}} g_1^4 L F_{3,0}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{54} \sum_{\mathbf{p}} g_1^4 L F_{4,-1}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{16}{405} \sum_{\mathbf{p}} g_1^4 L F_{5,-2}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{54} g_1^4 L F_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 L F_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{4}{135} g_1^4 L F_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{288} g_1^4 L F_{2,1,0}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{288} g_1^4 L F_{2,2,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{144} g_1^4 L F_{3,1,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{288} g_1^4 L F_{4,1,-2}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{288} g_1^4 L F_{2,1,0}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{288} g_1^4 L F_{2,2,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{144} g_1^4 L F_{3,1,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{288} g_1^4 L F_{4,1,-2}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{2592} g_1^4 L F_{2,1,0}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2592} g_1^4 L F_{2,2,-1}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{1296} g_1^4 L F_{3,1,-1}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2592} g_1^4 L F_{4,1,-2}[m_1, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{2592} g_1^4 L F_{2,1,0}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2592} g_1^4 L F_{2,2,-1}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{1296} g_1^4 L F_{3,1,-1}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2592} g_1^4 L F_{4,1,-2}[m_1, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{96} g_1^2 g_2^2 L F_{2,1,0}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{2,2,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^2 g_2^2 L F_{3,1,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{4,1,-2}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{96} g_1^2 g_2^2 L F_{2,1,0}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{2,2,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^2 g_2^2 L F_{3,1,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{4,1,-2}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{96} g_1^2 g_2^2 L F_{2,1,0}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{2,2,-1}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^2 g_2^2 L F_{3,1,-1}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{4,1,-2}[m_2, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{96} g_1^2 g_2^2 L F_{2,1,0}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{2,2,-1}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^2 g_2^2 L F_{3,1,-1}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{4,1,-2}[m_2, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{54} g_1^2 g_3^2 L F_{2,1,0}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 L F_{2,2,-1}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{27} g_1^2 g_3^2 L F_{3,1,-1}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 L F_{4,1,-2}[m_3, m_q^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{54} g_1^2 g_3^2 L F_{2,1,0}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 L F_{2,2,-1}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{27} g_1^2 g_3^2 L F_{3,1,-1}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 L F_{4,1,-2}[m_3, m_q^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{18} g_1^2 \bar{y}_d^{i4p} y_d^{i3p} L F_{2,1,0}[m_d^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{36} g_1^2 \bar{y}_d^{i4p} y_d^{i3p} L F_{2,2,-1}[m_d^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} + \\ & \quad \frac{1}{36} g_1^2 \bar{y}_d^{i4p} y_d^{i3p} L F_{3,1,-1}[m_d^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{18} g_1^2 \bar{y}_e^{i2p} y_e^{i1p} L F_{2,1,0}[m_e^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} - \\ & \quad \frac{1}{36} g_1^2 \bar{y}_e^{i2p} y_e^{i1p} L F_{2,2,-1}[m_e^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} - \frac{1}{36} g_1^2 \bar{y}_e^{i2p} y_e^{i1p} L F_{3,1,-1}[m_e^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} + \\ & \quad \frac{1}{144} g_1^4 L F_{2,1,0}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{288} g_1^4 L F_{3,1,-1}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^2 g_2^2 L F_{2,1,0}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{3,1,-1}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{144} g_1^4 L F_{2,1,0}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{288} g_1^4 L F_{3,1,-1}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^2 g_2^2 L F_{2,1,0}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 L F_{3,1,-1}[m_l^{$$